EXERCISE SHEET 8

TWO-SAMPLE TESTS

Hypothesis testing
In these exercises we will use the following procedure when testing hypotheses:

(1) State the parameter of interest.

(2) State the null hypothesis H_0.

(3) State the alternative hypothesis H_1.

(4) State the test statistic.

(5) State the conditions (critical value) for rejecting the null hypothesis.

(6) Calculate the value of the test statistic.

(7) Conclude on the test.

Exercise 10.1.1

Consider the hypothesis test $H_0 : \mu_1 = \mu_2$ against $H_1 : \mu 1 \neq \mu_2$ with std. deviations $\sigma_1 = 10$ and $\sigma_2 = 5$ (variances known). Suppose that sample sizes are $n_1 = 10$ and $n_2 = 15$, and that sample means are $\overline{x}_1 = 4.7$ and $\overline{x}_2 = 7.8$. Use significance level $\alpha = 0.05$.

(a) **Test the hypothesis and find the P-value.**

1. The parameter of interest here is the difference in means $\mu_1 - \mu_2 = \Delta$. Note that here $\Delta_0 = 0$.
2. $H_0 : \mu_1 - \mu_2 = 0$ or $\mu_1 = \mu_2$
3. $H_1 : \mu_1 - \mu_2 \neq 0$ or $\mu 1 \neq \mu_2$

4. The test statistic for this is:

$$z_0 = \frac{(\overline{x}_1 - \overline{x}_2) - \Delta_0}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

5. Reject the null hypothesis H_0 if $z_0 < -z_{\alpha/2} = -1.96$ or $z_0 > z_{\alpha/2} = 1.96$. (Can be found using the Matlab command `norminv(1 - α/2)`)

6. $\overline{x}_1 = 4.7 \quad \overline{x}_2 = 7.8$
   $\sigma_1 = 10 \quad \sigma_2 = 5$
   $n_1 = 10 \quad n_2 = 15$

$$z_0 = \frac{(4.7 - 7.8) - 0}{\sqrt{\frac{10^2}{10} + \frac{5^2}{15}}} = -0.9$$

7. **Conclusion**: Because the test statistic falls within $-1.96 < -0.9 < 1.96$, we do not reject the null hypothesis. There is not sufficient evidence to conclude that the two means differ at $\alpha = 0.05$.

We can then compute the P-value from the test statistic, using the cumulative distribution function for the normal distribution (`normcdf()` in Matlab) and complement probability. Note that we multiply by 2 because this is a two-tailed test.

$$\begin{aligned}
\text{P-value} &= 2 \cdot P(z > |z_0|) = 2 \cdot (1 - P(z \leq |z_0|)) \\
&= 2 \cdot (1 - \Phi(|z_0|)) = 2 \cdot (1 - \texttt{normcdf}(|z_0|)) \\
\text{P-value} &= 2 \cdot (1 - \Phi(0.9)) = 2(1 - 0.815950) = 0.368
\end{aligned}$$

(b) **Explain how the test could be conducted with a confidence interval.**
We could construct a confidence interval for the difference in means, with the same significance level as the test ($\alpha = 0.05$ resulting in a 95% CI), and then reject the null hypothesis if 0 is not contained within the interval. For completeness sake we will conduct the test in this manner and construct the CI:

$$(\overline{x}_1 - \overline{x}_2) - z_{\alpha/2}\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}} \leq \mu_1 - \mu_2 \leq (\overline{x}_1 - \overline{x}_2) + z_{\alpha/2}\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$$

$$(4.7 - 7.8) - 1.96\sqrt{\frac{10^2}{10} + \frac{5^2}{15}} \leq \mu_1 - \mu_2 \leq (4.7 - 7.8) + 1.96\sqrt{\frac{10^2}{10} + \frac{5^2}{15}}$$

$$-9.79 \leq \mu_1 - \mu_2 \leq 3.59$$

With 95% confidence the true difference in means is between -9.79 and 3.59. Because 0 is in this interval, we conclude there is no significant difference between the means, and fail to reject the null hypothesis.

Exercise 10.2.2

Consider the hypothesis test $H_0 : \mu_1 = \mu_2$ against $H_1 : \mu_1 \neq \mu_2$. Suppose that sample sizes are $n_1 = 15$ and $n_2 = 15$, that sample means are $\overline{x}_1 = 4.7$ and $\overline{x}_2 = 7.8$, and that the sample variances are $s_1^2 = 4$ and $s_2^2 = 6.25$.
Assume that $\sigma_1 = \sigma_2$ and that data are drawn from a normal distribution. Use significance level $\alpha = 0.05$.

(a) **Test the hypothesis and find the P-value.**

1. The parameter of interest here is the difference in means $\mu_1 - \mu_2 = \Delta$. Note that here $\Delta_0 = 0$.
2. $H_0 : \mu_1 - \mu_2 = 0$ or $\mu_1 = \mu_2$
3. $H_1 : \mu_1 - \mu_2 \neq 0$ or $\mu_1 \neq \mu_2$
4. The test statistic for this is:

$$t_0 = \frac{(\overline{x}_1 - \overline{x}_2) - \Delta_0}{s_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

   (Note that although this test is similar to the one in 10.1.1, in this problem the true variances are unknown, so we use a t-test instead of a z-test.)
5. Reject the null hypothesis H_0 if $t_0 < -t_{\alpha/2,n_1+n_2-2} = -2.048$ or $t_0 > t_{\alpha/2,n_1+n_2-2} = 2.048$. ($t_{0.025,28} = 2.048$ can be found using the Matlab command `tinv(`$1-\alpha/2$`,` n_1+n_2-2`)`)
6. $\overline{x}_1 = 4.7 \quad \overline{x}_2 = 7.8$
   $s_1^2 = 4 \quad s_2^2 = 6.25$
   $n_1 = 10 \quad n_2 = 15$
   To calculate the test statistic we first need to calculate the pooled sample variance s_p:

$$s_p = \sqrt{\frac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{n_1+n_2-2}}$$

$$s_p = \sqrt{\frac{14 \cdot 4 + 14 \cdot 6.25}{15+15-2}} = 2.26$$

$$t_0 = \frac{4.7 - 7.8}{2.26\sqrt{\frac{1}{15} + \frac{1}{15}}} = -3.75$$

7. **Conclusion**: Because the test statistic t_0 is below the critical value $-3.75 < -2.048$, we reject the null hypothesis at significance level $\alpha = 0.05$.

We can then compute the P-value from the test statistic, using the cumulative distribution function for the student-t distribution (`tcdf()` in Matlab) and complement probability.

Note that we multiply by 2 because this is a two-tailed test.

$$\begin{aligned}\text{P-value} &= 2 \cdot P(t > |t_0|) = 2 \cdot (1 - P(t \leq |t_0|)) \\ &= 2 \cdot (1 - \Phi(|t_0|)) = 2 \cdot (1 - \texttt{tcdf}(|t_0|, n_1 + n_2 - 2)) \\ \text{P-value} &= 2 \cdot (1 - \Phi(3.75)) = 2(1 - 0.9996) = 0.0008\end{aligned}$$

(b) **Explain how the test could be conducted with a confidence interval.**
We could construct a confidence interval for the difference in means, with the same significance level as the test ($\alpha = 0.05$ resulting in a 95% CI), and then reject the null hypothesis if 0 is not contained within the interval. For completeness sake we will conduct the test in this manner and construct the CI:

$$(\overline{x}_1 - \overline{x}_2) - t_{\alpha/2,n_1+n_2-2} \cdot s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}} \leq \mu_1 - \mu_2 \leq (\overline{x}_1 - \overline{x}_2) + t_{\alpha/2,n_1+n_2-2} \cdot s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$$

$$(4.7 - 7.8) - 2.048 \cdot 2.26 \sqrt{\frac{1}{15} + \frac{1}{15}} \leq \mu_1 - \mu_2 \leq (4.7 - 7.8) + 2.048 \cdot 2.26 \sqrt{\frac{1}{15} + \frac{1}{15}}$$

$$-4.79 \leq \mu_1 - \mu_2 \leq -1.41$$

With 95% confidence the true difference in means is between -4.79 and -1.41. Because 0 is not contained in this interval, we reject the null hypothesis that the means are equal.

Exercise 10.2.6

Two suppliers manufacture a plastic gear used in a laser printer. The impact strength of these gears measured in foot-pounds is an important characteristic. A random sample of $n_1 = 10$ gears from supplier 1 results in $\overline{x}_1 = 290$ and $s_1 = 12$. Another random sample of $n_2 = 16$ gears from supplier 2 results in $\overline{x}_1 = 321$ and $s_1 = 22$.

(a) **Is there evidence to support the claim that supplier 2 provides gears with higher mean impact strength? Use $\alpha = 0.05$ and assume that both populations are normally distributed but the variances are not equal. What is the P-value for this test?**

1. The parameter of interest here is the difference in mean impact strength $\mu_1 - \mu_2 = \Delta$, with $\Delta_0 = 0$.
2. $H_0 : \mu_1 - \mu_2 = 0$ or $\mu_1 = \mu_2$
3. $H_1 : \mu_1 - \mu_2 < 0$ or $\mu_1 < \mu_2$
4. The test statistic for this is:

$$t_0 = \frac{(\overline{x}_1 - \overline{x}_2) - \Delta_0}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

5. Reject the null hypothesis H_0 if $t_0 < -t_{\alpha,v}$, where $-t_{0.05,23} = -1.714$ since:

$$v = \frac{\left(s_1^2/n_1 + s_2^2/n_2\right)^2}{\frac{(s_1^2/n_1)^2}{n_1-1} + \frac{(s_2^2/n_2)^2}{n_2-1}} = 23.72$$

$$v \approx 23$$

$$t_{0.05,23} = \texttt{tinv(0.05,23)} = 1.1714$$

6. $\overline{x}_1 = 290 \quad \overline{x}_2 = 321$
   $s_1 = 12 \quad s_2 = 22$
   $n_1 = 10 \quad n_2 = 16$

$$t_0 = \frac{290 - 321}{\sqrt{\frac{12^2}{10} + \frac{22^2}{16}}} = -4.64$$

7. **Conclusion**: Because the test statistic t_0 is below the critical value $-4.64 < -1.714$, we reject the null hypothesis and conclude that supplier 2 provides gears with higher mean impact strength at significance level $\alpha = 0.05$.

We can then compute the P-value from the test statistic, using the cumulative distribution function for the student-t distribution (`tcdf()` in Matlab) and complement probability.

$$\begin{aligned}\text{P-value} &= P(t > |t_0|) = 1 - P(t \leq |t_0|) \\ &= 1 - \Phi(|t_0|) = 1 - 2 \cdot (1 - \texttt{tcdf}(|t_0|, v)) \\ \text{P-value} &= 1 - \Phi(4.64) = 1 - 0.99995 = 0.00005\end{aligned}$$

(b) **Does the data support the claim that the mean impact strength of gears from supplier 2 is at least 25 foot-pounds higher than that of supplier 1? Make the same assumptions as in part (a).**

1. The parameter of interest here is the difference in mean impact strength $\mu_2 - \mu_1 = \Delta$, with $\Delta_0 = 25$.
2. $H_0 : \mu_2 - \mu_1 = 25$
3. $H_1 : \mu_2 - \mu_1 > 25$ or $\mu_2 > \mu_1 + 25$
4. The test statistic for this is:

$$t_0 = \frac{(\overline{x}_2 - \overline{x}_1) - \Delta_0}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

5. Reject the null hypothesis H_0 if $t_0 > t_{\alpha,v}$, where $t_{0.05,23} = 1.714$.
6. $\overline{x}_1 = 290 \quad \overline{x}_2 = 321$
   $s_1 = 12 \quad s_2 = 22$

$n_1 = 10 \quad n_2 = 16$
$\Delta_0 = 25$

$$t_0 = \frac{(321 - 290) - 25}{\sqrt{\frac{12^2}{10} + \frac{22^2}{16}}} = 0.898$$

7. **Conclusion**: Because $t_0 = 0.898 < 1.714$, we fail to reject the null hypothesis. There is insufficient evidence to conclude that the mean impact strength of gears from supplier 2 is at least 25 ft-lb higher than gears from supplier 1, at significance level $\alpha = 0.05$.

(c) **Construct a confidence interval estimate for the difference in mean impact strength and explain how this interval could be used to answer the question posed regarding supplier-to-supplier differences.**
Using the information from part (a) and $t_{0.025,23} = 2.069$, we can construct a 95% confidence interval on the difference in mean strength $\mu_2 - \mu_1$ as follows:

$$(\overline{x}_2 - \overline{x}_1) - t_{0.025,23}\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}} \leq \mu_1 - \mu_2 \leq (\overline{x}_2 - \overline{x}_1) + t_{0.025,23}\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

$$31 - 2.048 \cdot 2.069\sqrt{\frac{12^2}{10} + \frac{22^2}{16}} \leq \mu_1 - \mu_2 \leq 31 + 2.069\sqrt{\frac{12^2}{10} + \frac{22^2}{16}}$$

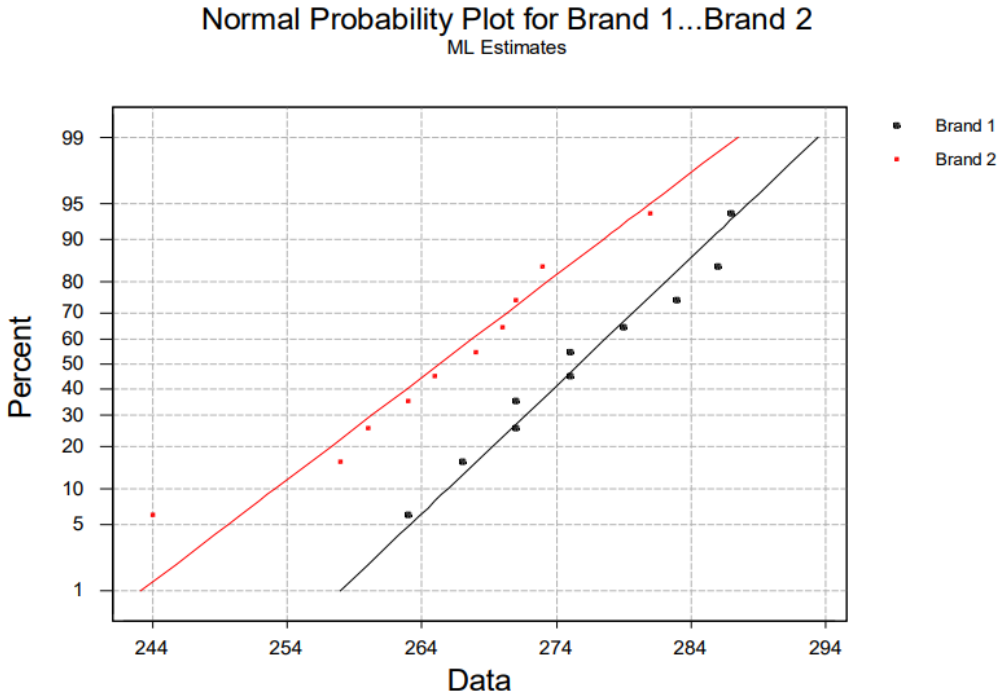
$$17.175 \leq \mu_1 - \mu_2 \leq 44.825$$

Because 0 is not contained in this interval, we conclude that supplier 2 provides gears with higher mean impact strength than supplier 2 with 95% confidence.

Exercise 10.2.11

The overall distance traveled by a golf ball is tested by hitting the ball with the Iron Byron, a mechanical golfer with a swing that is said to emulate the distance hit by the legendary champion Byron Nelson. The randomly selected balls of two different brands are tested and the overall distance is measured. **The (Matlab friendly) data follows:**
Brand1 = [275 286 287 271 283 271 279 275 263 267];
Brand2 = [258 244 260 265 273 281 271 270 263 268];

(a) **Is there evidence that overall distance is approximately normally distributed? Is an assumption of equal variances justified?**
The data is examined in a normal probability plot, shown below. The data appear to be normally distributed and the variances appear to be approximately equal. The slopes of the lines on the normal probability plots are almost the same.



(b) **Test the hypothesis that both brands of ball have equal overall distance. Use $\alpha = 0.05$. What is the P-value?**

1. The parameter of interest here is the difference in mean impact strength $\mu_1 - \mu_2 = \Delta$, with $\Delta_0 = 0$.
2. $H_0 : \mu_1 - \mu_2 = 0$ or $\mu_1 = \mu_2$
3. $H_1 : \mu_1 - \mu_2 \neq 0$ or $\mu_1 \neq \mu_2$
4. The test statistic for this is:
$$t_0 = \frac{(\overline{x}_1 - \overline{x}_2) - \Delta_0}{s_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$
5. Reject the null hypothesis H_0 if $t_0 < -t_{\alpha/2,n_1+n_2-2} = -2.101$ or $t_0 > t_{\alpha/2,n_1+n_2-2} = 2.101$. ($t_{0.025,18} = 2.101$ can be found using the Matlab command `tinv(`$1-\alpha/2$`,` n_1+n_2-2`)`)
6. $\overline{x}_1 = 275.7 \quad \overline{x}_2 = 265.3$
$s_1 = 8.03 \quad s_2 = 10.04$
$n_1 = 10 \quad n_2 = 10$
To calculate the test statistic we first need to calculate the pooled sample variance s_p:
$$s_p = \sqrt{\frac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{n_1+n_2-2}}$$
$$s_p = \sqrt{\frac{9 \cdot 8.03^2 + 9 \cdot 10.04^2}{10+10-2}} = 9.09$$

$$t_0 = \frac{275.7 - 265.3}{9.09\sqrt{\frac{1}{10} + \frac{1}{10}}} = 2.558$$

7. **Conclusion**: Because the test statistic t_0 is above the critical value $2.558 > 2.101$, we reject the null hypothesis at significance level $\alpha = 0.05$. The data supports the claim that means are different.

We can then compute the P-value from the test statistic. Note that we multiply by 2 because this is a two-tailed test.

$$\text{P-value} = 2 \cdot (1 - \Phi(|t_0|)) = 2 \cdot (1 - \texttt{tcdf}(|t_0|, n_1 + n_2 - 2))$$

$$\text{P-value} = 2 \cdot (1 - \Phi(2.558)) = 2(1 - 0.9901) = 0.0198$$

(c) **Construct a 95% two-sided CI on the mean difference in overall distance for the two brands of golf balls.**
We can construct the 95% two-sided CI in the same way we did in 10.2.2 (b):

$$(\overline{x}_1 - \overline{x}_2) - t_{\alpha,v} s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}} \leq \mu_1 - \mu_2 \leq (\overline{x}_1 - \overline{x}_2) + t_{\alpha,v} s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$$

$$(275.7 - 265.3) - 2.101(9.09)\sqrt{\frac{1}{10} + \frac{1}{10}} \leq \mu_1 - \mu_2 \leq (275.7 - 265.3) + 2.101(9.09)\sqrt{\frac{1}{10} + \frac{1}{10}}$$

$$1.86 \leq \mu_1 - \mu_2 \leq 18.94$$

Note that 0 is not contained in this interval, which fits the conclusion we found in (b).

Exercise 10.4.1

The manager of a fleet of automobiles is testing the two brands of radial tires and assign one tire of each brand at random to the two rear wheels of eight cars and runs the cars until the cars until the tires wear out. **The (Matlab friendly) data (in kilometers) follows:**
Brand1 = [36925 45300 36240 32100 37210 48360 38200 33500];
Brand2 = [34318 42280 35500 31950 38015 47800 37810 33215];

Find a 99% confidence interval on the difference in mean life. Which brand would you prefer based on this calculation?
We have $n = 8$ samples. Using Matlab we can compute the sample mean and sample std. deviation of the difference in means with `mean(Brand1-Brand2)` and `std(Brand1-Brand2)`

$$\overline{d} = 868.375$$

$$s_d = 1290.033$$

$$\text{where } d_i = \text{Brand1}_i - \text{Brand2}_i$$

We can find the critical value from the level of significance $\alpha = 0.01$ by using the command `tinv(1 - 0.01/2,n-1)`; $t_{\alpha/2,7} = 3.499$ Finally we can construct the 99% confidence interval

$$\overline{d} - t_{\alpha/2,n-1}\left(\frac{s_d}{\sqrt{n}}\right) \leq \mu_d \leq \overline{d} + t_{\alpha/2,n-1}\left(\frac{s_d}{\sqrt{n}}\right)$$

$$868.375 - 3.499\left(\frac{1290}{\sqrt{8}}\right) \leq \mu_d \leq 868.375 + 3.499\left(\frac{1290}{\sqrt{8}}\right)$$

$$-727.46 \leq \mu_d \leq 2464.21$$

Because this confidence interval contains zero, there is no significant difference between the two brands of tire at a 1% significance level.

Exercise 10.4.3

An article in the *Journal of Aircraft* described a new equivalent plate analysis method formulation that is capable of modelling aircraft structures such as cranked wing boxes and that produces results similar to the more computationally intensive finite element analysis method. Natural vibration frequencies for the cranked wing box structure are calculated using both methods, and results for the first seven natural frequencies (in Cycle/s) follow:
(in a Matlab friendly format)
FiniteElement = [14.58 48.52 97.22 113.99 174.73 212.72 277.38];
EquivalentPlate = [14.76 49.10 99.99 117.53 181.22 220.14 294.80];

(a) **Does the data suggest that the two methods provide the same mean value for for natural vibration frequency? Use $\alpha = 0.05$. Find the P-value.**

1. The parameter of interest is the mean difference in natural vibration frequencies, μ_d where $d_i = \text{FiniteElement}_i - \text{EquivalentPlate}_i$.
2. $H_0 : \mu_d = 0$
3. $H_1 : \mu_d \neq 0$
4. The test statistic is
$$t_0 = \frac{\overline{d}}{s_d/\sqrt{n}}$$
5. Reject the null hypothesis if $t_0 < -t_{0.025,6}$ or $t_0 > t_{0.025,6}$ where $t_{0.025,6} = 2.447$ for $\alpha = 0.05$
6. $\overline{d} = -5.49$
$s_d = 5.924$
$n = 7$
$$t_0 = \frac{-5.49}{5.924/\sqrt{7}} = -2.45$$

7. **Conclusion**: Because $-2.45 < -2.447$, we reject the null hypothesis. The two methods have different mean values for natural vibration frequency at the 0.05 level of significance.

(b) **Find a 95% confidence interval on the mean difference between the two methods.**

Using the same critical value as earlier $t_{0.025,6} = 2.447$, we construct the CI as:

$$\overline{d} - t_{\alpha/2,n-1}\left(\frac{s_d}{\sqrt{n}}\right) \leq \mu_d \leq \overline{d} + t_{\alpha/2,n-1}\left(\frac{s_d}{\sqrt{n}}\right)$$

$$-5.49 - 2.447\left(\frac{5.924}{\sqrt{7}}\right) \leq \mu_d \leq -5.49 + 2.447\left(\frac{5.924}{\sqrt{7}}\right)$$

$$-10.969 \leq \mu_d \leq -0.011$$

With 95% confidence, the mean difference between the natural vibration frequency from the equivalent plate method and the finite element method is between -10.969 and -0.011 cycles.